Solutions & gases

Choose units

1 Mass concentration

c in g/dm³ = mass in g / solution volume in dm³. Mass = cV.

250 cm³ = **0.250 dm³.** 5.0 g in 0.250 dm³ → **20 g/dm³.**

Same solute, double final volume → half concentration.

2 Molar concentration

n = cV with c in mol/dm³ and V in dm³. Rearrange: c = n/V.

0.200 mol/dm³ × 0.150 dm³ = 0.0300 mol. For NaOH, mass = 0.0300 × 40 = **1.20 g.**

mol/dm³ × molar mass → g/dm³.

Use reaction ratios

3 Titration route

c × V	moles	ratio

Convert → compare → convert back

Take mean of concordant titres, excluding rough trial. Convert both volumes to dm³.

Known moles = known c × V. Use balanced equation ratio. Unknown concentration = unknown moles / its volume.

4 A 1: 2 example

H_2SO_4 + 2NaOH → Na_2SO_4 + $2H_2O$.

20.0 cm³ acid at 0.100 mol/dm³ → 0.00200 mol acid → 0.00400 mol NaOH.

If NaOH volume is 25.0 cm³, c = 0.00400/0.0250 = **0.160 mol/dm³.**

Convert gas amounts

5 Gas volume at RTP

One mole occupies **24 dm³** at 20 °C and 1 atmosphere. Equal moles → equal gas volumes at same conditions.

V = 24n; n = V/24, with V in dm³. 1 dm³ = 1,000 cm³.

4.4 g CO_2 / 44 = 0.10 mol → **2.4 dm³.**

6 Gas equation ratios

$2H_2$ + O_2 → $2H_2O$: 60 cm³ H_2 needs 30 cm³ O_2 at the same conditions.

For mass to gas: mass → moles → equation ratio → gas moles → volume.

Do not apply the gas-volume rule to a solid or liquid; identify limiting gas if both amounts are given.